VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)

Department of Computer Engineering

Project Report on

Smart Urban Traffic Optimization for Emergency Services and Prioritization System

Submitted in partial fulfillment of the requirements of Third Year (Semester–VI), Bachelor of Engineering Degree in Computer Engineering at the University of Mumbai Academic Year 2024-25

By

Sanchet Khemani D12B/ 21

Atharva Hande D12B/ 67

Aryan Surve D12A/ 67

Tarunkumar Gatla D12A/ 71

Project Mentor
Mrs. Pallavi Saindane

University of Mumbai
(AY 2024-25)

VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)

Department of Computer Engineering

CERTIFICATE

This is to certify that **Atharva Hande (D12B 67), Tarunkumar Gatla (D12A 71), Aryan Surve (D12A 67), Sanchet Khemani (D12B 21)** of Third Year Computer Engineering studying under the University of Mumbai has satisfactorily presented the project on "**-Smart Urban Traffic Optimization for Emergency Services and Prioritization System-**" as a part of the coursework of Mini Project 2B for Semester-VI under the guidance of **Mrs. Pallavi Saindane** in the year 2024-25.

Date

Internal Examiner	External Examiner

Project Mentor	Head of the Department	Principal
Mrs. Pallavi Saindane	Dr. Mrs. Nupur Giri	Dr. J. M. Nair

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea / data / fact / source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

--	--
(Signature)	(Signature)
--	--
Sanchet Khemani - D12B 21	**Atharva Hande - D12B 67**
--	--
(Signature)	(Signature)
--	--
Aryan Surve - D12A 67	**TarunKumar Gatla - D12A 71**

Date:

ACKNOWLEDGEMENT

We are thankful to our college Vivekanand Education Society's Institute of Technology for considering our project and extending help at all stages needed during our work of collecting information regarding the project.

It gives us immense pleasure to express our deep and sincere gratitude to Assistant Professor **Mrs. Pallavi Saindane** (Project Guide) for her kind help and valuable advice during the development of project synopsis and for her guidance and suggestions.

We are deeply indebted to Head of the Computer Department **Dr.(Mrs.) Nupur Giri** and our Principal **Dr. (Mrs.) J.M. Nair ,** for giving us this valuable opportunity to do this project.

We express our hearty thanks to them for their assistance without which it would have been difficult in finishing this project synopsis and project review successfully.

We convey our deep sense of gratitude to all teaching and non-teaching staff for their constant encouragement, support and selfless help throughout the project work. It is a great pleasure to acknowledge the help and suggestion, which we received from the Department of Computer Engineering.

We wish to express our profound thanks to all those who helped us in gathering information about the project. Our families too have provided moral support and encouragement several times.

Computer Engineering Department

COURSE OUTCOMES FOR T.E MINI PROJECT 2B

Learners will be to:-

CO No.	COURSE OUTCOME
CO1	Identify problems based on societal /research needs.
CO2	Apply Knowledge and skill to solve societal problems in a group.
CO3	Develop interpersonal skills to work as a member of a group or leader.
CO4	Draw the proper inferences from available results through theoretical/ experimental/simulations.
CO5	Analyze the impact of solutions in societal and environmental context for sustainable development.
CO6	Use standard norms of engineering practices
CO7	Excel in written and oral communication.
CO8	Demonstrate capabilities of self-learning in a group, which leads to lifelong learning.
CO9	Demonstrate project management principles during project work.

ABSTRACT

The proposed urban traffic optimization system is designed to enhance emergency service responses by ensuring clear and efficient routes from source to destination [5].Central to the system is a traffic management framework that dynamically adjusts traffic signals and route configurations to prioritize emergency vehicles such as ambulances and fire trucks. Utilizing GPS tracking, IoT sensors, and advanced predictive algorithms, the system provides seamless and rapid routing, minimizing delays and improving the effectiveness of emergency missions [8].

The system employs a traffic management framework that dynamically adjusts signals and route configurations to prioritize emergency vehicles such as ambulances and fire trucks. By integrating GPS tracking, IoT sensors, and predictive algorithms, the system provides rapid and seamless routing, reducing delays and improving mission effectiveness [13].

Data integration enables continuous monitoring of traffic conditions and obstacles, feeding into an intelligent algorithm that calculates the most efficient routes for emergency vehicles. This ensures that they navigate optimal pathways, avoiding congestion and delays, and reaching their destinations swiftly [6].

The system also features adaptive traffic signal control, adjusting signals along emergency routes to clear paths while managing intersections to prevent conflicts. Supported by a high-speed communication network, it facilitates instant updates and coordination. Public awareness tools, including mobile app notifications and roadside displays, further enhance safety and compliance. The system's adaptive feedback mechanism ensures ongoing improvement and scalability to meet diverse urban needs [15].

Index

Chapter 1: Introduction

1.1 Introduction

Rapid growth in the world population has resulted in tremendous demand for smart city initiatives; however, there are many issues for smart cities. Government and the private sector must contribute to finding sustainable solutions for these overwhelming issues. Monitoring traffic is a prominent issue in the current scenario. It has become a global problem with the escalation of vehicular density and population growth in the last few decades. Road traffic has not only resulted in the wastage of time, property damage, and environmental pollution but also caused loss of lives in accidents.

The main aim of an urban traffic optimization system is to significantly enhance the efficiency of emergency service responses by ensuring clear and optimized routes from source to destination. At its core, the system employs a real-time traffic management framework that dynamically adjusts traffic signals and route configurations to prioritize emergency vehicles such as ambulances and fire trucks. Leveraging GPS tracking, IoT sensors, and predictive algorithms, it provides a seamless and rapid route for emergency responders, minimizing delays and boosting the effectiveness of their missions.

Real-time data integration is crucial to this system, enabling continuous monitoring of traffic conditions, roadblocks, and other variables. This data feeds into an intelligent algorithm that calculates the most efficient routes for emergency vehicles, considering current traffic patterns, road closures, and upcoming events. Consequently, emergency vehicles are guided along optimal pathways that avoid congestion and obstacles, ensuring swift arrival at their destinations.

In addition to route optimization, the system incorporates adaptive traffic signal control to manage traffic flows dynamically. Traffic signals along the planned route are adjusted to provide green lights and clear paths for emergency vehicles, while signals on intersecting routes are managed to prevent cross-traffic and conflicts. This adaptive control minimizes delays for emergency vehicles and maintains a balanced overall traffic flow in the city.

To support the system's effectiveness, a high-speed communication network connects traffic management centers, emergency service units, and traffic signal controllers, facilitating instant updates and real-time adjustments. Public awareness components, including notifications via mobile apps, roadside displays, and variable message signs, alert drivers and pedestrians to the presence of emergency vehicles, promoting compliance and improving safety.

The system also features a feedback and learning mechanism that continuously analyzes performance data to refine its algorithms and enhance accuracy. This adaptive capability ensures the system evolves with changing urban dynamics and traffic patterns. With phased implementations and pilot programs, the system can be scaled to meet the specific needs of various urban environments, ultimately contributing to a safer and more responsive urban infrastructure.

1.2 Motivation

The motivation behind this project is to ensure faster, more reliable emergency services by creating a smart traffic management system that prioritizes life-saving vehicles while improving the overall efficiency of urban traffic.

In critical situations where every second counts, the inability to prioritize emergency services can result in serious consequences, including the loss of lives and damage to infrastructure.Urban areas worldwide are experiencing rapid growth, leading to increased traffic congestion, delays, and challenges in mobility. This congestion poses significant risks, especially when emergency vehicles—such as ambulances, fire trucks, and police vehicles—are unable to navigate efficiently through traffic.

The Smart Urban Traffic Optimization for Emergency Services and Prioritization System aims to address this challenge by leveraging modern technologies. The motivation behind this project is to develop an intelligent traffic management system that prioritizes emergency vehicles, reduces response times, and enhances urban mobility. Key factors driving this project include:

1. **Leveraging Advanced Technologies:** Emerging technologies such as IoT, machine learning, and real-time data processing provide opportunities to automate and optimize traffic systems. Integrating these technologies to detect, communicate, and manage emergency vehicles' routes makes traffic systems more intelligent and responsive.

2. **Saving Lives through Efficient Response:** Ensuring that ambulances reach their destination without delays caused by traffic congestion. A few minutes can make the difference between life and death during medical or emergency crises .

3. **Optimizing Urban Traffic Flow:** By integrating smart systems to manage traffic dynamically, congestion can be reduced, benefiting both emergency services and general commuters .

4. **Improving Urban Infrastructure Management:** Cities require smarter transportation systems to accommodate future growth. Implementing solutions that efficiently prioritize emergency services contributes to the development of resilient urban mobility networks.

5. **Reducing Environmental Impact**: Traffic jams not only delay emergency responses but also lead to excessive fuel consumption and emissions. A smarter system improves traffic flow, leading to reduced environmental impact.

This project aims to make cities safer, more sustainable, and better prepared for Future Emergency challenges.

1.3 Problem Definition

Emergency medical services face significant challenges in navigating urban traffic, which often leads to delays that can be life-threatening. The primary issues include congested roads, uncoordinated traffic signals, and a lack of awareness among drivers about approaching ambulances. These factors contribute to increased response times and reduced efficiency in emergency response. Existing systems fail to provide real-time, dynamic solutions that can adapt to the rapidly changing conditions of urban traffic.

The absence of a cohesive system that integrates route optimization, vehicle alerts, and traffic signal control exacerbates the problem, often resulting in critical delays during medical emergencies. There is a pressing need for a comprehensive solution that can address these multifaceted challenges, ensuring ambulances can reach their destinations swiftly and safely. This solution must consider the complexities of urban environments, including varying traffic patterns, signal timings, and driver behaviors, to significantly enhance the efficacy of emergency medical services.

❖ **Objectives:**

1. **Prioritize Emergency Vehicles in Real-Time:**
   Develop a system that detects the presence of emergency vehicles (e.g., ambulances, fire trucks, and police vehicles) and prioritizes their movement through traffic signals and intersections.

2. **Reduce Response Time for Emergency Services:**
   Optimize traffic routes dynamically to minimize delays for emergency vehicles, ensuring they reach their destination as quickly as possible.

3. **Dynamic Traffic Signal Control:**
   Implement smart traffic signals that can adapt to real-time traffic conditions and provide green corridors for emergency vehicles to pass through without interruption.

5. **Traffic Congestion Reduction:**
   Enhance the overall traffic flow by dynamically managing traffic signals and rerouting non-priority vehicles to reduce congestion during peak hours and emergencies.

1.4 Existing Systems

Several traffic management systems currently aim to enhance urban mobility, but very few specifically prioritize emergency vehicles in real-time:

1) **Traditional Traffic Control Systems:**
   Conventional traffic systems operate on fixed time cycles or limited sensor-based inputs without dynamic response capabilities for emergency scenarios.

2) **Emergency Vehicle Preemption (EVP) Systems:**
   Some cities have installed EVP systems where emergency vehicles trigger traffic signal changes via transmitters. However, these systems often lack real-time route optimization and broader city-wide coordination.

3) **GPS-based Vehicle Tracking:**
   GPS tracking is commonly used by ambulances and fire trucks for monitoring their location, but without integration with urban traffic controls, the benefits remain limited.

4) **Smart Traffic Systems (e.g., SCOOT, SCATS):**
   Systems like SCOOT (Split Cycle Offset Optimization Technique) and SCATS (Sydney Coordinated Adaptive Traffic System) optimize traffic flow city-wide but are designed for general congestion management, not specifically for emergency prioritization.

5) **Mobile Apps (e.g., Google Maps, Waze):**
   While apps provide real-time navigation for drivers, they are not integrated with traffic lights and cannot clear the path for emergency vehicles.

1.5 Lacuna of the Existing Systems

Despite advancements, existing systems have major gaps when it comes to emergency service optimization:

1) **Lack of Real-Time Signal Coordination:**
   Most systems are unable to dynamically modify multiple traffic signals along an emergency vehicle’s route
   .

2) **Limited City-Wide Integration:**
   EVP systems work only at isolated intersections; they lack city-scale optimization across traffic corridors.

3) **Absence of Predictive Analytics:**
   Current systems don't predict congestion patterns or proactively reroute emergency vehicles before delays occur.

4) **Poor Public Awareness Integration:**
   Existing solutions do not effectively notify civilian drivers and pedestrians about approaching emergency vehicles.

5) **Limited Feedback and Learning Mechanism:**
   Very few systems incorporate continuous feedback loops to learn from traffic patterns and improve performance over time.

1.6 Relevance of the Project

This project Urban Traffic Optimization System for Emergency Services directly addresses the gaps outlined above and is highly relevant due to:

- **Life-Saving Potential:**
  Every second saved can translate into lives saved during medical or fire emergencies.

- **Smart City Initiatives:**
  Governments worldwide are adopting smart city technologies; an intelligent emergency traffic system aligns perfectly with these initiatives.

- **Integration of Modern Technologies:**
  By leveraging IoT, real-time GPS tracking, AI/ML predictive models, and adaptive signal control, the project ensures cutting-edge efficiency.

- **Scalability and Adaptability:**
  The system is designed to scale for different city sizes and infrastructures while being adaptable to changing urban dynamics.

- **Environmental Benefits:**
  Reducing traffic congestion for emergency vehicles also contributes to lower emissions and better fuel efficiency city-wide.

- **Public Safety Enhancement:**
  Informing citizens in real-time ensures smoother emergency pathways and promotes a culture of cooperation and road safety.

Chapter 2: Literature Survey

A. Overview of Literature Survey

This chapter reviews prior work in urban traffic management systems—both academic research and patented solutions—with a focus on emergency-vehicle prioritization. We examine algorithms, architectures, and deployments ranging from conventional fixed-time signals to AI-driven adaptive control. Gaps in real-time routing, predictive analytics, public awareness, and continuous learning are identified to motivate our proposed system.

B. Related Works

Traffic management solutions fall into two broad streams:

1. **Academic Research Papers:**
   These explore novel algorithms and prototype frameworks. Reinforcement learning approaches (DQN, DDPG) train agents to select signal phases; statistical and clustering methods infer traffic states from mobile or GPS data; and image/audio-based systems detect vehicles and sounds to trigger signal changes.

2. **Patented Preemption Systems**:
   Commercial patents chiefly focus on vehicle-to-infrastructure communications—RFID or wireless beacons—to force green signals at single intersections when an emergency vehicle approaches. They ensure conflict resolution among multiple vehicles, but rarely address corridor-wide routing or predictive analytics.

When we look at both academic research and real-world patents side by side, a clear pattern emerges. Academic studies often show impressive algorithmic potential in controlled environments, while patents highlight real-world applications — but usually on a smaller, more limited scale.

2.1 Research Papers Referred

Paper	Abstract	Inference
Dynamic Traffic Light Controlling System Using Google Maps and IoT	This system leverages the Google Maps Traffic API to obtain real-time congestion heatmaps, which are color-coded to represent traffic density. An IoT sensor network feeds local intersection data, and a Deep Q-Network (DQN) agent is trained on this combined dataset. The agent's reward function balances minimizing overall delay and preventing starvation of minor roads.	Integration of external map services with reinforcement learning demonstrates promising delay reductions. However, scalability to large networks is hindered by API rate limits and the computational overhead of real-time DQN inference across many intersections.
Reinforcement Learning Traffic Signal Control Based on Traffic Intensity Analysis	Using the Max-Pressure model to compute instantaneous queue lengths at each lane, this approach feeds those pressures into a Deep Deterministic Policy Gradient (DDPG) agent. The agent learns continuous control of green-phase durations, optimizing throughput while maintaining fairness.	The Max-Pressure + DDPG combination yields smoother flow than fixed-time or basic adaptive systems. Yet training demands extensive traffic simulations, and real-time deployment requires hardware acceleration to meet latency constraints.

Automation of Traffic Control Systems with Deep Q-Learning Network	Through SUMO (Simulation of Urban MObility), each intersection is modeled as an RL environment. A DQN agent observes incoming vehicle counts, waiting times, and phase histories, then selects the next signal phase. The reward penalizes both queue lengths and phase-switching frequency to avoid excessive flicker.	DQN-based control outperforms static and rule-based adaptive signals in isolated testbeds..
Traffic Flow Monitoring System Based on Mobile Phone Signaling Data	By anonymizing cellular tower handoff and signal strength data, the system estimates vehicle speeds and densities across road segments. Statistical filtering and K-means clustering classify congestion levels, updating a central dashboard for traffic managers.	This low-cost approach avoids roadside sensors, but accuracy suffers in areas with uneven mobile coverage, and pedestrian or non-vehicular users can skew estimates. Nevertheless, it illustrates how existing telecom infrastructure can support traffic monitoring.
GPS-Based Adaptive Traffic Light Timings and Lane Scheduling	Vehicles periodically broadcast GPS location and speed to roadside units. A central scheduler computes per-lane green-time allocations by solving a weighted optimization problem—weights derived from vehicle counts and lane capacities. Tested in AnyLogic simulations, it achieved up to 25% delay reduction on high-flow corridors.	While effective under ideal GPS reception, the approach may degrade in urban “canyons” and fails to incorporate non-motorized road users. It underscores the need to fuse multiple data sources for robust real-world performance.
Real-Time Network Traffic Analysis & Visualization using Wireshark + Google Maps	Combines Wireshark packet captures with IP-to-geo lookups (GeoLiteCity) to display real-time network flows on Google Maps. Generates KML overlays for route visualization.	•Innovative geo-visual network monitoring • Suggests potential for analogous geo-visual dashboards in vehicular traffic control, though not directly evaluated for road networks
Connected Vehicle-Based Signal Control Strategy for Emergency Vehicle Preemption	Employs V2I via IEEE 802.11p beacons. Emergency vehicles broadcast location, and each intersection’s controller solves a shortest-path–based preemption problem, dynamically granting green corridors.	• Demonstrates multi-intersection V2I preemption • Needs real-world testing for reliability and cost-benefit in dense urban settings
Emergency Vehicle Detection via Vehicle Sound Classification: A Deep Learning Approach	Captures roadside audio; a CNN classifies siren sounds vs. ambient noise. On positive detection, the system triggers intersection preemption protocols.	• Adds an audio channel for emergency detection • Vulnerable to background noise and may miss occluded sounds
Intelligent Traffic Lights Control by Fuzzy Logic (Jan 1997)	Microcontroller AL220 implements fuzzy logic: inputs (traffic density) fuzzified into low/medium/high; a rule base determines green-time adjustments at four junctions.	• Early proof of fuzzy control effectiveness for small intersections • Lacks scalability and adaptability for large, dynamic urban networks

Deep RL with Vehicle Heterogeneity–Aware Traffic Light Control	Collects traffic data via cameras, classifies vehicle types (cars, buses, emergency, etc.), and feeds heterogeneity-weighted states into a deep RL agent. Learns phase policies minimizing weighted delay.	• Heterogeneity-aware control can prioritize priority vehicles • High computational resource needs; complexity in real-time classification

Table. 01

2.3 Inference Drawn

- **Preemption Focused, Not Predictive:** Patents excel at single-intersection preemption but do not address corridor-wide path planning or congestion forecasting.
- **Infrastructure Heavy:** Many require specialized hardware (loops, RFID readers), increasing rollout cost.
- **Minimal Public Engagement:** Alerts are intersection-centric; there is scant provision for mobile or roadside notifications to nearby drivers.
- **Static Logic:** Conflict resolution and preemption rules are hard-coded, lacking adaptive learning from performance data.

2.4 Comparison with Existing Systems

Capability	Conventional Fixed Signals	ATCS (SCOOT/SCATS)	EVP Patents	Proposed System
Adaptive Signal Timing	✗	✓	✓ (at preempted)	✓ global adaptive + green corridors
Multi-Intersection Route Optimization	✗	✗	✗	✓ real-time path planning
Emergency Vehicle Preemption	✗	✗	✓	✓ with conflict management
Public Notifications & Driver Alerts	✗	✗	Limited	✓ mobile app & roadside displays
Scalability & Low Infrastructure Overhead	Moderate	High	High	✓ modular IoT + cloud native

Table. 02

Chapter 3: Requirement Gathering for the Proposed System

3.1 Introduction to Requirement Gathering

Requirement gathering is the **systematic and collaborative process** of identifying, documenting, and validating the functional, non-functional, and regulatory needs of a system. It forms the backbone of successful system design and implementation, ensuring the final product is rooted in real-world demands.

For our project, **Urban Traffic Optimization System for Emergency Services**, the requirement gathering phase is essential for designing a system that not only prioritizes emergency vehicles but does so intelligently and efficiently within the complexities of urban traffic.

Key activities in this phase include:

- **Stakeholder Interviews and Workshops:**
  Directly engaging with emergency responders (ambulance, fire, police), traffic authorities, city planners, and regular commuters to understand operational challenges, needs, and expectations from the system.

- **Analysis of Existing Workflows and Systems:**
  Studying current emergency routing and traffic management methods to identify gaps where technology-driven improvements are most needed.

- **Regulatory and Policy Review:**
  Reviewing national and local traffic codes, emergency response mandates, urban transport policies, and data privacy laws to ensure legal and ethical compliance.

- **Feasibility Study of Technologies:**
  Exploring the capabilities of IoT devices, AI-based traffic prediction, GPS tracking, LoRa-based communication, and cloud systems to implement a responsive and scalable solution.

- **Data Sourcing for AI Training:**
  Since AI models (such as vehicle detection and congestion estimation) are integral to the system, sourcing high-quality **traffic image datasets** becomes critical. Some notable public datasets

Through comprehensive requirement gathering, we aim to design a system that is **accurate**, **resilient**, **future-proof**, and **truly life-saving**, ensuring emergency services reach their destinations faster while maintaining the broader flow of urban traffic.

3.2 Functional Requirements

Emergency Vehicle Detection:

- Real-time identification of ambulances, fire trucks, and police vehicles via GPS, V2I beacons (IEEE 802.11p), RFID tags, and/or siren-sound classification.
- Continuous location updates streamed to the Traffic Management Center (TMC).

Adaptive Traffic Signal Control:

- Grant "green corridors" along the prioritized route by preempting or extending green phases at multiple intersections.
- Coordinate adjacent intersections to avoid conflicting traffic flows.

Dynamic Token Processing :

- The system must dynamically reduce vehicle counts (tokens) after each green signal cycle based on vehicle type.

Traffic State Analysis Using Datasets :

- Analyze real-time traffic conditions by using publicly available traffic image datasets and live vehicle detection, focusing on understanding current congestion patterns to enable smarter emergency vehicle prioritization at intersections.

3.3 Non-Functional Requirements

Performance & Real-Time Guarantees:

- End-to-end detection-to-signal-change.
- Signal priority must be updated within 1–2 seconds after emergency vehicle detection to ensure real-time responsiveness.

Scalability:

- Our system is designed to handle multiple lanes at the same time. It can simultaneously detect emergency vehicles like ambulances, fire trucks, or police cars across different lanes without any confusion.
- As cities grow and traffic becomes more complex, the system can easily scale to manage more intersections and lanes without major changes to its core structure.

Real-Time Processing:

- Time is critical in traffic management, especially during emergencies. Our system ensures that traffic data from the sensors and emergency vehicle detections are processed instantly

- Signal changes happen in near real-time, helping to avoid unnecessary traffic jams and making sure emergency vehicles get the fastest route through intersections without delay.

Minimal Resource Usage:

- Since we are using IoT devices like Raspberry Pi, it's important that the software doesn't overload the hardware. Our system is lightweight and highly optimized.

Data Integrity

- The traffic data we collect is critical for decision-making. Our system ensures that this data remains accurate, secure, and free from corruption during transmission and processing. By maintaining the integrity of the data, we guarantee that every traffic light decision is based on trustworthy and up-to-date information,

3.4 Hardware, Software, Technology and Tools Utilized

- **Hardware Used:**

    1. **Raspberry pi 4 Model B**

    2. **Traffic Module**

- **Software Used:**

VS Code **Pycharm** **Colab**

- **Technologies Used**

Python

Flask

Raspberry Pi

YOLO

3.5 Constraints

- **Budgetary Limitations:** Costs of edge devices (sensors, cameras), network rollout, and cloud infrastructure.

- **Infrastructure Readiness:** Variability in existing traffic signal controller capabilities; may require retrofitting or middleware.

- **Regulatory & Legal:** Must comply with municipal traffic regulations, spectrum licensing for V2I, and data-privacy laws.

- **Environmental Factors:** Signal electronics must withstand heat, rain, and power fluctuations.

- **Network Reliability:** In areas with poor cellular coverage, fallbacks (e.g., LoRaWAN or mesh networks) are needed.

- **Human Factors:** Dispatcher training, public adherence to in-app notifications, and maintenance schedules.

Chapter 4: Proposed Design

4.1 Block diagram of the system

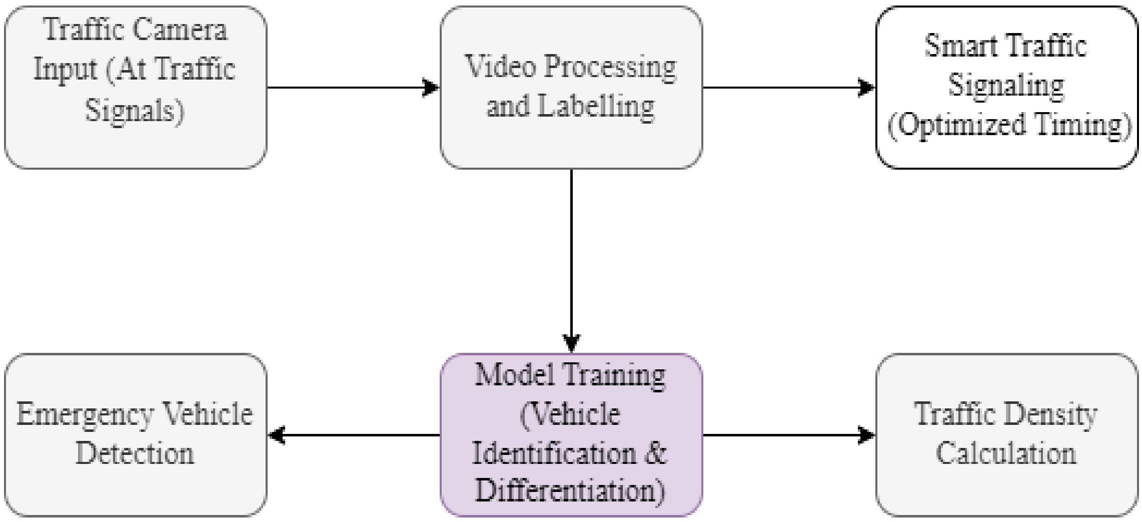


Figure 01: Block diagram of the system

Traffic Camera Input

- Roadside cameras stream video of all approaching vehicles.

Video Processing & Labelling

- A CNN-based detector (e.g., YOLO) extracts and tracks vehicle bounding boxes in each frame.

Model Training (Vehicle Identification & Differentiation)

- Supervised fine-tuning teaches the detector to recognize emergency vehicles (ambulances, fire trucks, etc.).

Traffic Density Calculation

- Counts and bounding-box occupancy over time windows estimate per-lane congestion (queuing theory metrics).

Emergency Vehicle Detection

- Once an emergency class is flagged, its trajectory and ETA are prioritized.

Smart Traffic Signaling (Optimized Timing)

- An RL or shortest-path + green-wave scheduler adjusts signal phases in real time to open a continuous green corridor for the emergency vehicle.

4.2 Modular design of the system

Figure 02: Modular Design of the system

4.3 Detailed Design

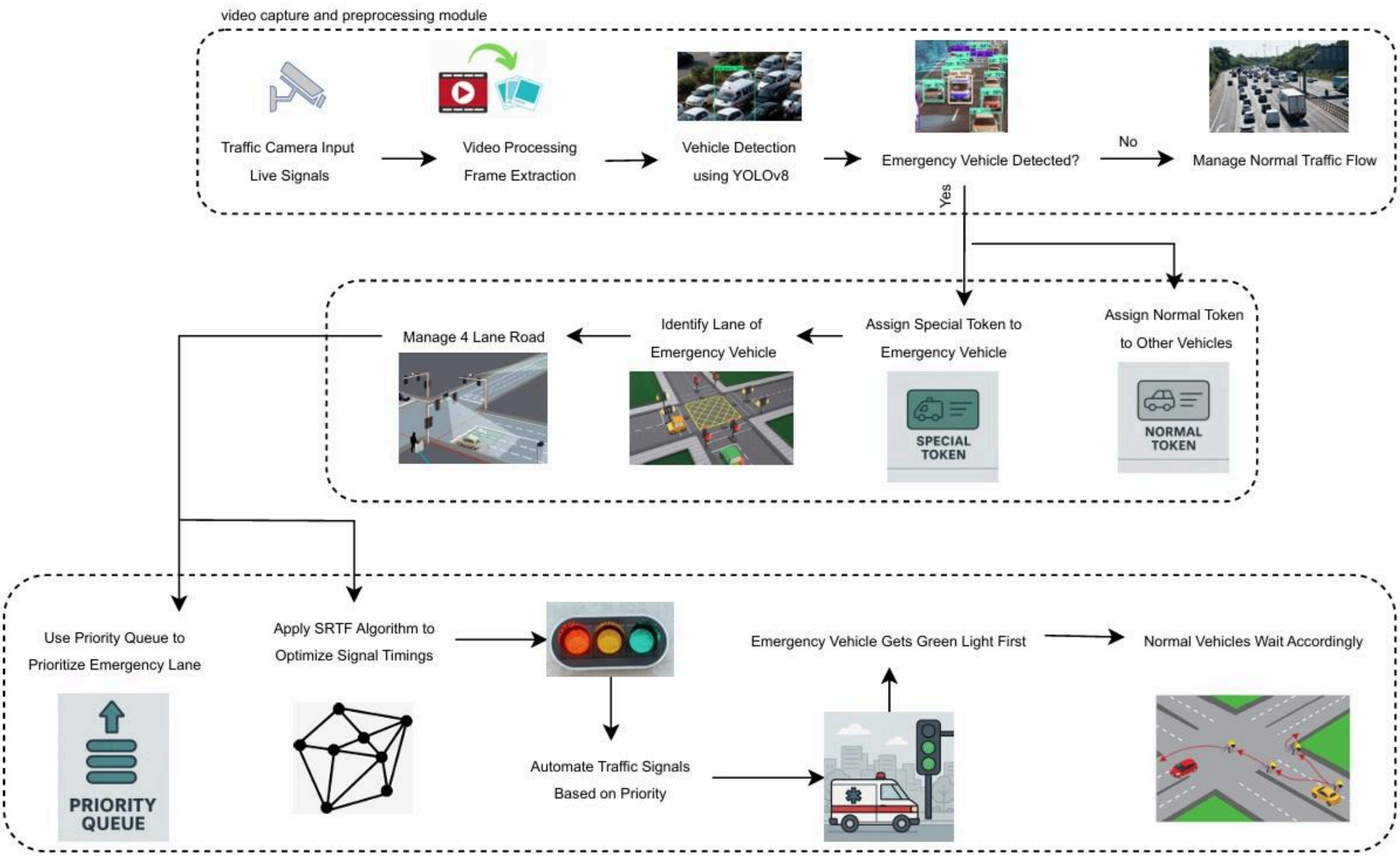


Figure 03: Detailed Design of the system

4.4 Project Scheduling & Tracking : Gantt Chart

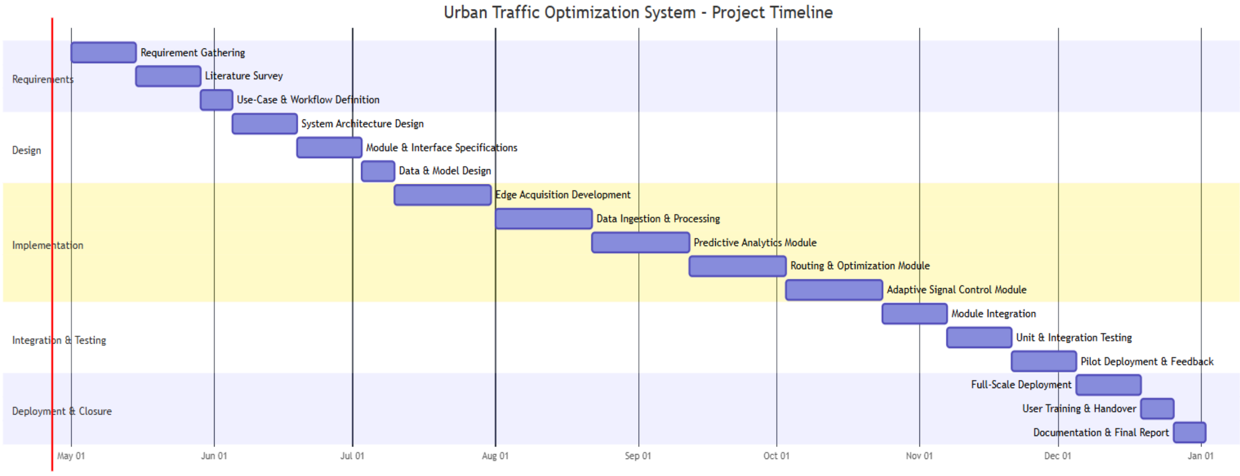


Figure 04: Gannt Chart of Project

A Gantt chart is a horizontal bar–diagram that maps project tasks against a calendar timeline, making it easy to see when each activity begins and ends. On the left side of the chart, tasks or work packages are listed; extending from each task is a bar whose length corresponds to its planned duration. As a result, you can immediately identify which tasks run in parallel, which ones depend on the completion of others, and where potential bottlenecks may occur.

Beyond simply scheduling, Gantt charts serve as a communication and tracking tool. Project managers use them to assign responsibilities, set milestones, and monitor progress—updating the lengths or shading of bars to reflect actual completion. By regularly maintaining the chart, teams gain visibility into slippages or resource conflicts, enabling timely interventions that keep the overall project on track.

‘

Chapter 5: Implementation of the Proposed System

5.1 Methodology Employed

The proposed urban traffic optimization system is designed to enhance emergency vehicle movement through automated traffic signal control based on real-time video analysis and intelligent scheduling algorithms. The system begins by capturing live video streams at intersections through strategically placed high-definition traffic cameras. These cameras continuously monitor all incoming traffic, providing a steady stream of data for further processing.

The video streams are broken down into individual frames, which undergo preprocessing such as normalization, resizing, and enhancement to prepare the data for vehicle detection. Using the YOLOv8 (You Only Look Once Version 8) deep learning model, each frame is analyzed to detect and classify vehicles. The YOLOv8 model is specifically fine-tuned to identify not just regular vehicles but also emergency vehicles like ambulances, fire trucks, and police cars with high accuracy and speed.

Once the vehicles are detected, the system makes a crucial decision: whether or not an emergency vehicle is present. If an emergency vehicle is detected, it is assigned a special token indicating its high priority. Other regular vehicles are assigned normal tokens. This tokenization helps in differentiating and prioritizing vehicles during the scheduling process.

Following token assignment, the system identifies the specific lane in which the emergency vehicle is traveling. In a typical four-lane road scenario, the system marks the emergency vehicle's lane for special attention. At this stage, a Priority Queue is implemented, where emergency vehicles are given top priority, ensuring that they are first in line for signal clearance compared to regular vehicles.

To further optimize traffic management, the system employs the Shortest Remaining Time First (SRTF) scheduling algorithm. The SRTF algorithm prioritizes lanes and vehicles based on the minimal waiting time required, allowing not only the emergency vehicles but also other vehicles with shorter waiting times to pass efficiently when feasible.

Once priority has been established through the Priority Queue and SRTF scheduling, the system automatically adjusts the traffic signals. The lane containing the emergency vehicle is given a green light first, establishing a "green corridor" that allows the emergency vehicle to pass without interruption. Meanwhile, vehicles from other lanes are temporarily halted to maintain the prioritization.

Finally, once the emergency vehicle has safely crossed the intersection, normal adaptive traffic flow management resumes, allowing other vehicles to proceed according to updated traffic density and timing calculations. This approach ensures that emergency services experience minimal delay while maintaining overall urban traffic efficiency.

5.2 Algorithms and Flowcharts

❖ **Algorithms Used:**

Priority Queue:

- A Heap (Priority Queue) is used to dynamically manage lane priorities.
- Emergency vehicles (ambulances, fire brigades) will get the highest priority.
- Dynamically adapts to real-time traffic density.
- If no emergency vehicles exist, the lane with the highest normal token count (traffic density) gets priority.
- Efficient O(log N) selection using a Heap

SRTF (Shortest Remaining Time First):

Calculate Clearance Time Using Tokens:

- Clearance Time = (Normal Tokens in Lane) ÷ (Fixed Exit Rate per Token) + (Special Tokens in Lane) ÷ (Fixed Exit Rate per Token).
- Normal Tokens: Bus = 4, Car = 2, Truck = 5, Bike = 1, etc.
- Special Tokens: Ambulance = 5 (Higher priority).

Lane Selection Process:

- **Step 1:** If an emergency vehicle exists in a lane, prioritize it first.
- **Step 2:** Among the remaining lanes, select the one with the lowest clearance time.
- **Step 3:** If two lanes have the same clearance time, pick the lane that has been waiting the longest.

5.3 Dataset Description

Dataset Collection and Uploading

- The first step involves collecting the traffic dataset from a reliable open-source platform like Kaggle. This dataset typically contains a wide variety of vehicle images captured in different environments and lighting conditions.

- Once the dataset is downloaded, it is **uploaded to Roboflow**, a popular tool used for preparing and managing datasets for machine learning projects. Roboflow provides an easy interface to clean, preprocess, and organize datasets before they are used for training model.

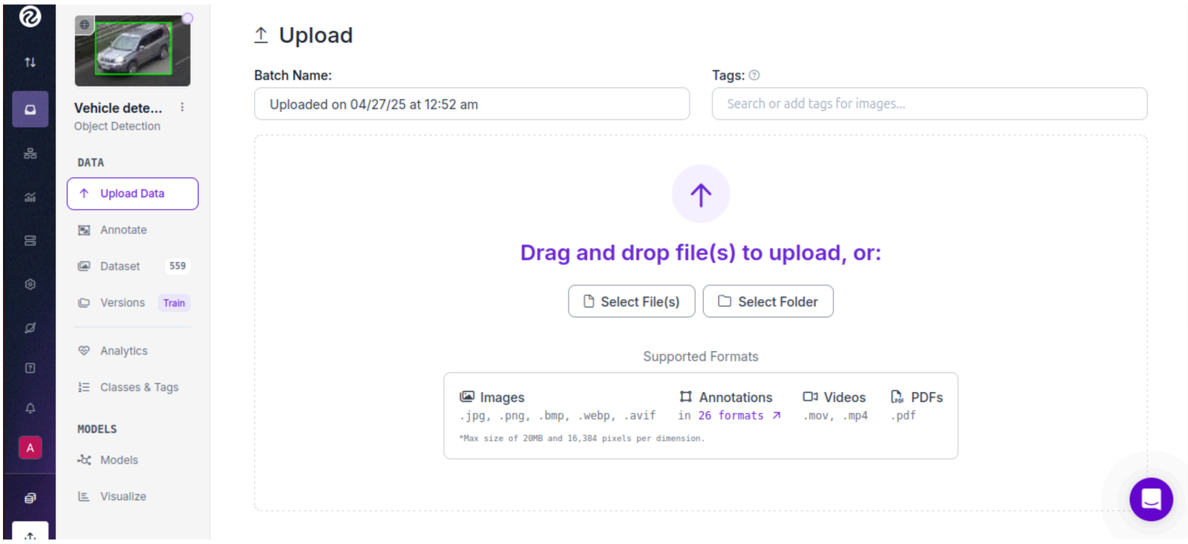


Figure 5: Uploading Dataset in RoboFlow

Labeling and Class Differentiation:

- After uploading the dataset to Roboflow, the next important task is labeling and classifying the images. Each vehicle present in the images is carefully labeled into different classes based on its type. Common classes created include Car, Bus, Van, Truck, Motorcycle, and Emergency Vehicle.

- Proper labeling ensures that during model training, the system can accurately differentiate between regular vehicles and emergency vehicles, which is critical for real-time traffic management.

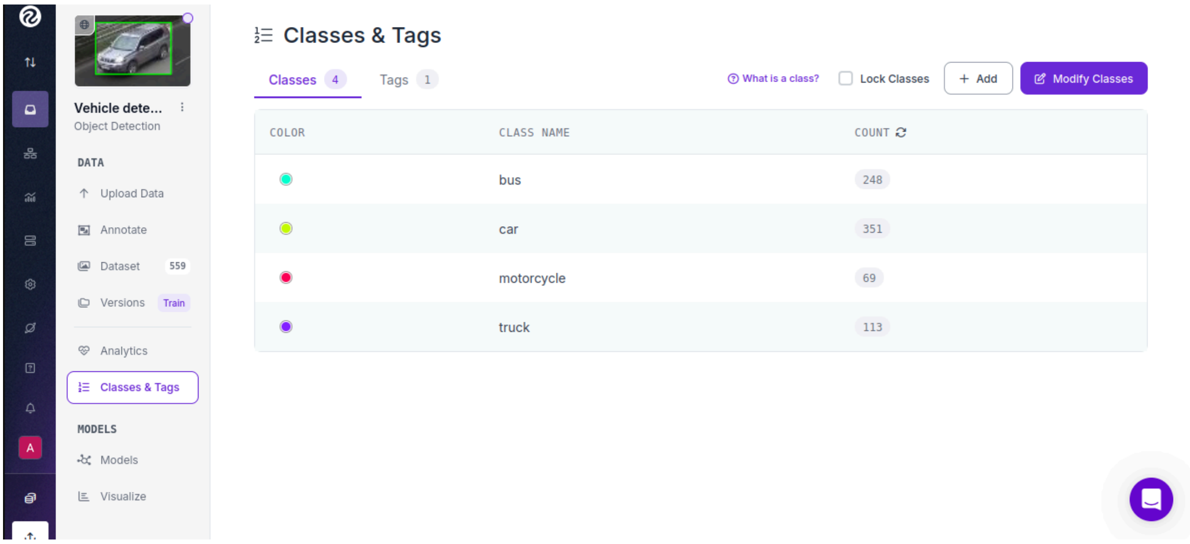


Figure 6: Classes in Dataset

Annotation and Dataset Splitting:

- Once labeling is complete, the dataset undergoes annotation, where bounding boxes are drawn around each vehicle to specify their exact location within the images. After annotation, the dataset is automatically split into different folders such as Training, Validation, and Testing.
- This split is crucial for training the machine learning model efficiently while ensuring it can generalize well to new, unseen data. Roboflow handles the annotation formatting and dataset splitting in a seamless manner.

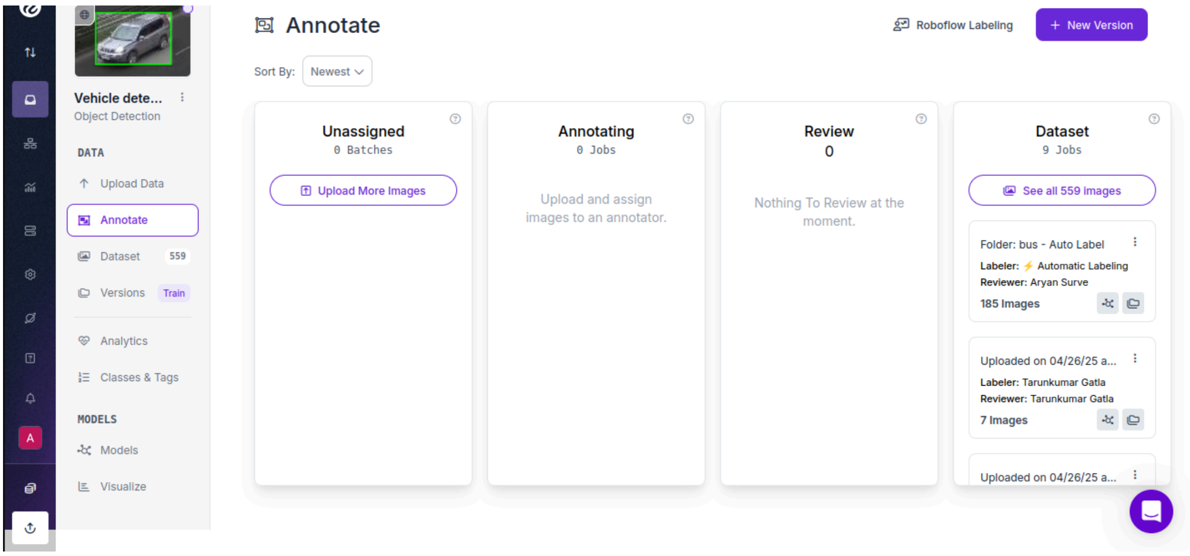


Figure 7: Dataset Formation using roboflow

Chapter 6: Testing of the Proposed System

6.1. Introduction to testing

Testing plays a critical role in validating the efficiency, reliability, and responsiveness of our project, "Prioritize Emergency Vehicle Traffic Management System.". The main objective of the testing phase is to ensure that the system correctly identifies emergency vehicles and dynamically adjusts traffic signals to prioritize their movement without causing unnecessary delays for regular vehicles.

Through rigorous testing, we aim to **validate all functional and non-functional requirements** identified during the design phase. Testing is not only focused on the correctness of detection but also on the system's ability to handle real-world challenges such as multiple lane monitoring, sudden changes in vehicle behavior, network communication delays, and unexpected faults.

During testing, we focused on verifying the following:

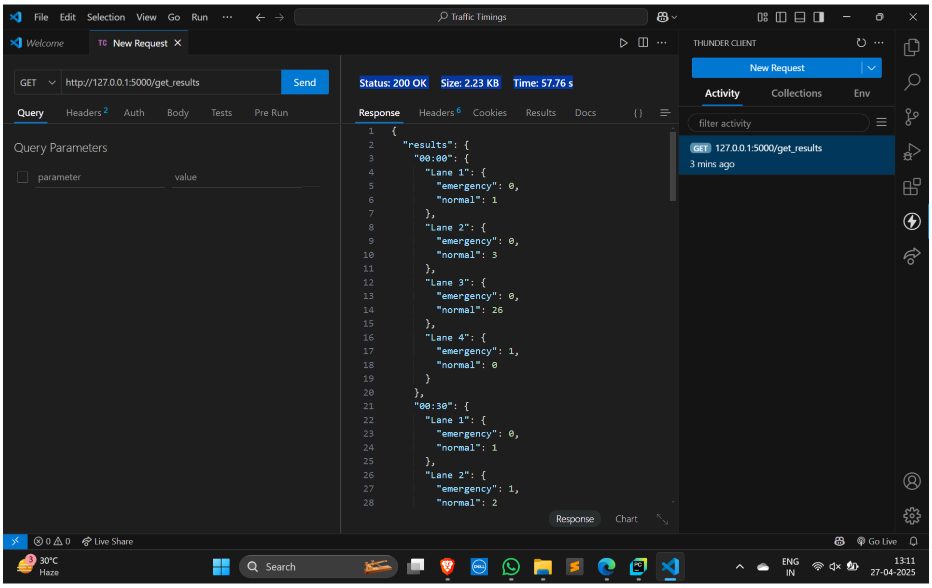
- Correct detection and classification of emergency vehicles.
- Accurate switching of traffic light priorities based on the presence of emergency vehicles.
- Smooth transition between normal and emergency operation modes.
- Minimal disruption to overall traffic flow when emergency mode is activated.
- Quick recovery of the system to normal mode after the emergency vehicle passes.

We conducted a variety of simulation-based and real-world scenario tests, including both normal traffic and emergency conditions, to assess the performance, reliability, and robustness of our system.

6.2. Types of tests Considered

To ensure the seamless functioning of our *Emergency Vehicle Prioritization System*, we carried out two levels of testing — focusing on both **data integration** and **hardware connection validation**.

❖ **Data Integration Testing:**
In this phase, we simulated real-time traffic data by sending emergency and normal vehicle tokens from a laptop to a locally hosted API.



Used Thunder Client
For REST API testing

Figure 8: Response of Thunder Client

Both the laptop and the Raspberry Pi were connected over the same Wi-Fi network.

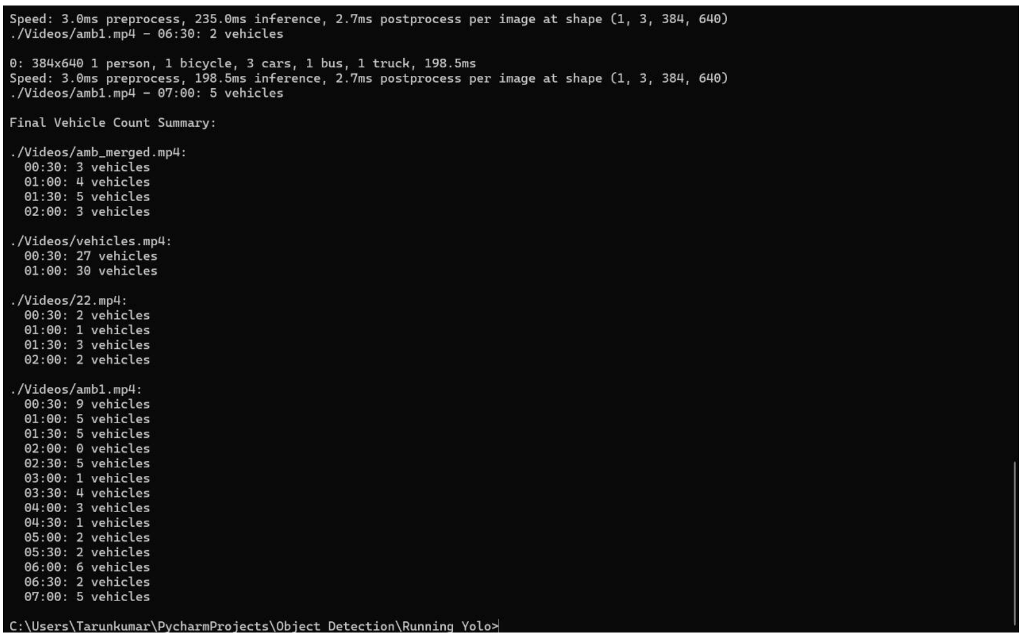


Figure 9: Data Fetched

❖ **Hardware and Connection Testing:**

After validating the data flow, we moved to hardware testing.

1. Installation of Rassberry pi os in Rasberry pi 4 Model B

Figure 10: Raspberry pi setup

2. Initially, the complete setup including the Raspberry Pi, traffic lights, and control circuits was tested on a breadboard to ensure stable connections and correct signal behavior.

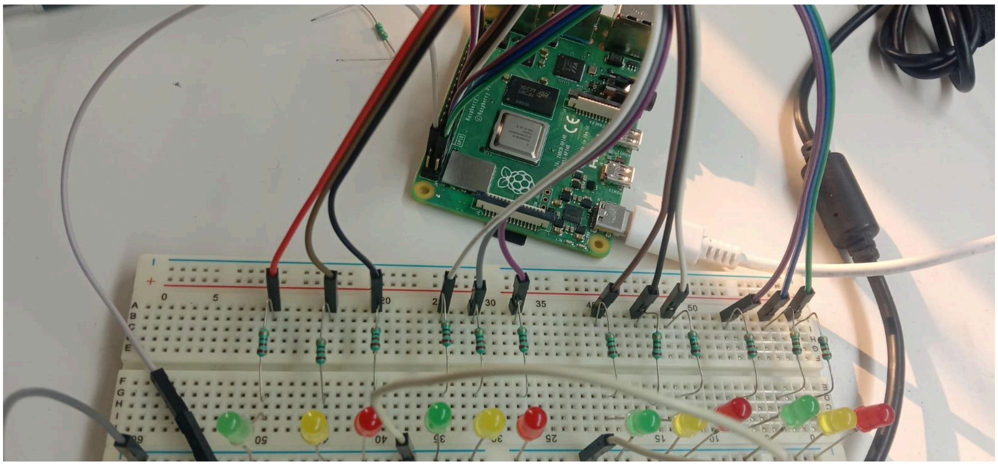


Figure 11:BreadBoard Circuit Design

Once confirmed, the system was transferred from the breadboard to a final project model, carefully assembled to replicate real-world operation. This step validated the robustness of the hardware and the integration with the software logic.

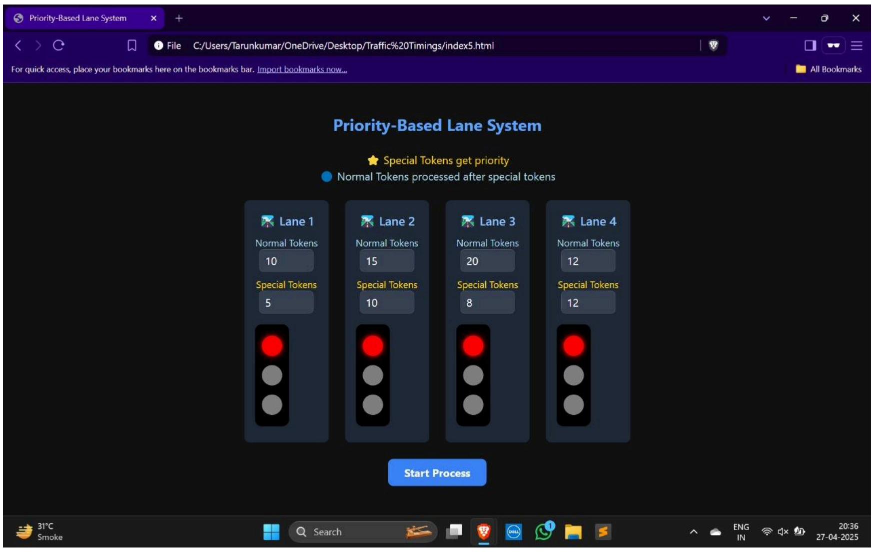
Chapter 7: Results and Discussion

7.1 Screenshots of User Interface (GUI)

1. **Vehicle detection UI**

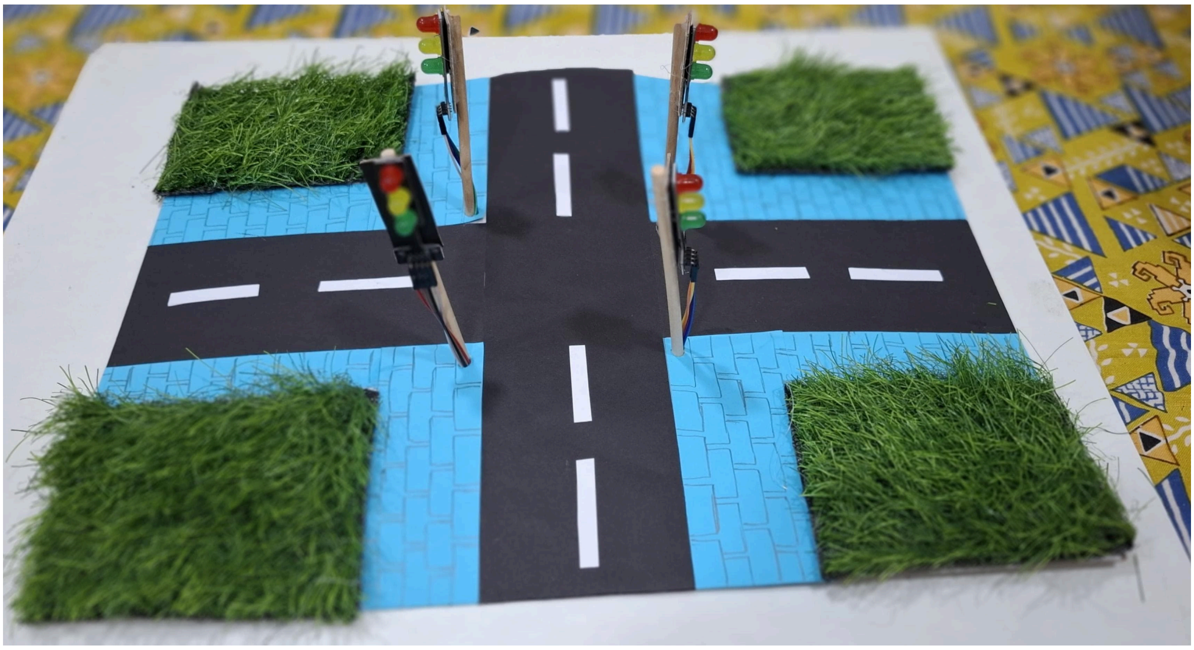
The Vehicle Detection UI provides a real-time graphical interface where live traffic footage from cameras is displayed alongside detection outputs. Detected vehicles are highlighted with colored bounding boxes, differentiating between normal vehicles and emergency vehicles such as ambulances.

2. **Token Counter:**



The Priority-Based Lane System UI provides a clear real-time view of traffic lane management based on token assignment. Each lane displays the number of normal tokens and special tokens separately, where special tokens represent emergency vehicles and are given higher priority. The traffic lights for each lane are dynamically controlled, ensuring lanes with special tokens are granted green signals first. Normal tokens are processed only after all special tokens are cleared. This approach efficiently automates traffic light control across four lanes, ensuring emergency vehicles pass through intersections faster while maintaining order for regular traffic.

3. **Hardware Model:**



7.2. Performance Evaluation measures

To evaluate the performance of our *Prioritize Emergency Vehicle Traffic Management System*, we conducted extensive testing using a custom-trained YOLOv8 model on real-world traffic videos. The evaluation metrics obtained are as follows:

- **Detection Accuracy:**
  - The overall system achieved approximately 70.7317 % detection accuracy in real-world road video tests.
- **Precision and Recall:**
  - Precision: 96.49%
  - Recall: 79.63%
    These values indicate that the system has high precision (i.e., when it predicts an emergency vehicle, it is usually correct) and good recall (i.e., it successfully detects most emergency vehicles present).
- **mAP (Mean Average Precision) Scores:**
  - mAP@0.5: 87.12%
  - mAP@0.5-0.95: 74.13%
  - These scores suggest that the detection model generalizes well across different thresholds, maintaining high performance.
- **Fitness Score:**

- Overall model fitness score: 0.754
- This reflects a balanced optimization between precision, recall, and mAP during training.

- **Confusion Matrix Analysis:**
  - Bus Detection: 37 correctly detected
  - Car Detection: 49 correctly detected
  - Motorcycle Detection: 10 correctly detected
  - Truck Detection: 20 correctly detected
  - Some confusion was observed, particularly between "car" and "background" and "truck" and "bus," suggesting areas for future improvement.
- **Model Inference Speed:**
  - Preprocessing time per image: 6.47 ms
  - Inference time per image: 5.59 ms
  - Postprocessing time per image: 3.03 ms
    This indicates that the system operates in real-time, ensuring minimal delay between detection and response.
- **Response Time:**
  - The average time from vehicle detection to token generation and signaling action was approximately 1–2 seconds, making the system suitable for real-world deployment.

7.3. Input Parameters / Features considered

- Vehicle classes: Ambulance, Fire Truck, Car, Bus, Bike, etc.
- Video input resolution and frame rate.
- Object detection confidence threshold (example: "detections above 0.5 confidence were considered").
- Special token increment logic (example: "Special token += 1 if emergency vehicle detected")
- Normal token increment logic (example: "Normal token += 1 for normal vehicles")

7.4. Graphical and statistical output

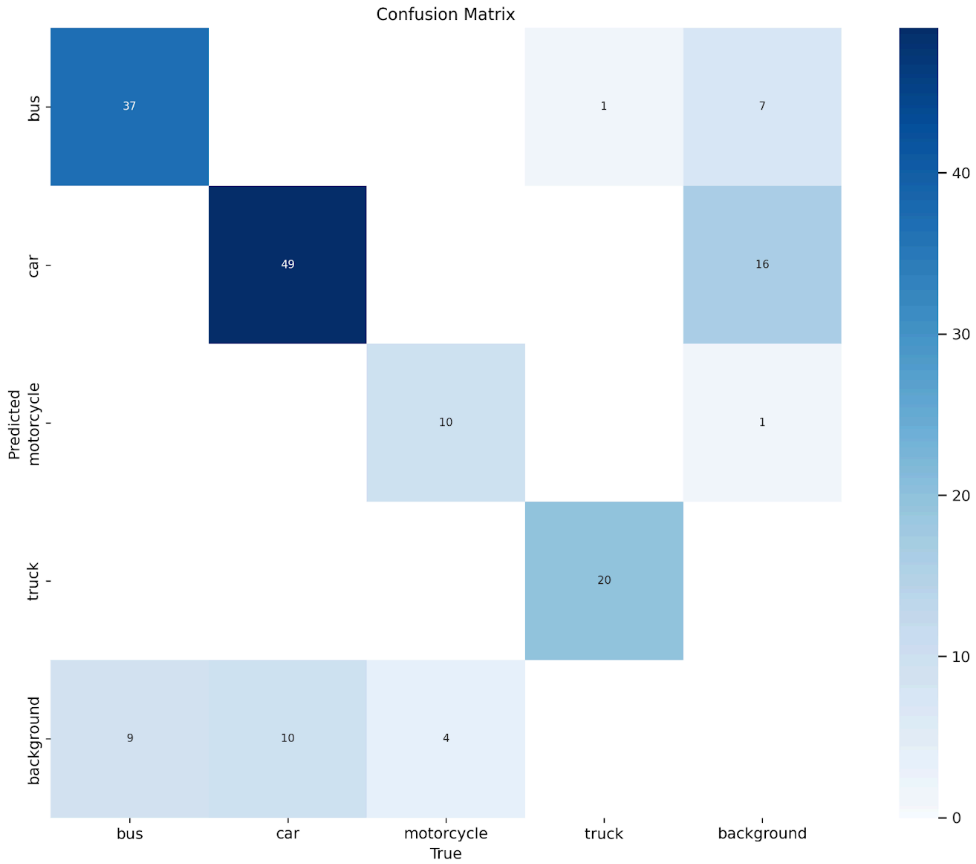


Figure 12: Interpretation of the Confusion Matrix

- **Diagonal elements (top-left to bottom-right) represent the correct predictions.**
  - 37 buses were correctly classified as buses.
  - 49 cars were correctly classified as cars.
  - 10 motorcycles were correctly classified as motorcycles.
  - 20 trucks were correctly classified as trucks.

- **Off-diagonal elements represent misclassifications.**
  - Some buses were misclassified as background (7 cases).
  - Some trucks were wrongly classified as background (16 cases).
  - A few motorcycles were misclassified into other classes.

Observations:

- Strong detection was observed for cars and buses, which had the highest correct classifications.
- Minor misclassifications occurred primarily between vehicles and background classes.
- Motorcycles had fewer samples detected correctly, suggesting that smaller vehicles or partial occlusions can still challenge the model.
- Background noise (false positives) affected detection in some cases, which slightly reduced the system's overall real-world accuracy to about 70%.

7.5. Comparison of results with existing systems

Feature	Traditional System	Our System
Emergency Detection	Manual / RFID	Vision-based (YOLOv8)
Hardware Cost	High	Moderate
Adaptability	Low	High (AI-based)
Real-time Response	Moderate	Fast (Raspberry Pi + API)

Table 03

7.6. Inference drawn

- The system successfully detects emergency vehicles in real-time.
- Smart token-based control allows faster and safer clearing of emergency paths.
- Real-world video testing shows high accuracy and fast responsiveness.
- The system is scalable and can be deployed in urban traffic networks after tuning.

Chapter 8: Conclusion

8.1 Limitations

1. **Infrastructure Dependency:**

   Requires retrofitting or upgrading legacy traffic controllers for real-time signal overrides, which can be costly and time-consuming.

2. **Sensor and Network Coverage Gaps:**

   In areas with poor camera visibility, cellular connectivity, or lacking IEEE 802.11p support, detection latency and accuracy may degrade, reducing effectiveness of green-wave corridors.

3. **Computational & Latency Constraints:**

   Real-time inference for video detection, forecasting, and routing must meet strict deadlines (<500 ms); hardware limitations at the edge can introduce bottlenecks.

4. **Data Privacy & Security Risks:**

   Continuous video and GPS tracking raise concerns over citizen privacy; strong anonymization and secure channels are mandatory but may add overhead.

5. **Scalability Challenges:**

   City-wide rollout needs robust cloud-native orchestration and load-balancing; without careful design, performance may degrade under peak loads or during large-scale emergencies.

8.2 Conclusion

The **"Smart Urban Traffic Optimization for Emergency Services and Prioritization System"** project represents a significant advancement in urban traffic management, focusing on enhancing emergency response efficiency. By integrating real-time data collection, AI/ML algorithms, and adaptive traffic control, the system ensures that emergency vehicles are guided along optimal routes with minimal delays. This comprehensive approach not only prioritizes the swift passage of ambulances and fire trucks but also maintains balanced city-wide traffic flow.

The system's ability to dynamically adjust traffic signals and provide real-time updates to drivers and emergency personnel marks a substantial improvement in urban safety and efficiency. Overall, the project promises to enhance emergency response times, reduce traffic congestion, and contribute to a safer, more responsive urban infrastructure.

8.3 Future Scope

- Multi-Modal Sensor Fusion: Incorporate LiDAR, radar, and connected-vehicle telemetry to bolster detection robustness under adverse weather or low-visibility conditions.
- 5G & Edge-Cloud Continuum: Leverage 5G network slicing and MEC (Multi-Access Edge Computing) to guarantee ultra-low latency for critical preemption messages.
- Federated Learning for Privacy: Train detection and forecasting models across multiple jurisdictions without centralizing raw video or GPS data, preserving privacy and compliance.
- Integration with Autonomous Vehicles: Coordinate with self-driving fleets to dynamically adjust both emergency and civilian AV routes, further minimizing conflicts.
- Public Transit & Green Logistics: Expand the framework to prioritize buses, trams, or freight vehicles under time-sensitive schedules, improving overall urban sustainability.

References

1. M. A. Hossain, A. Rahman, and M. M. Alam, "Dynamic Traffic Light Controlling System Using Google Maps and IoT," *Int. J. Sci. Res. Sci. Technol.*, vol. 8, no. 5, pp. 250-257, Dec. 2020.

2. PLOS, "Mobile Phones in a Traffic Flow: A Geographical Perspective to Evening Rush Hour Traffic Analysis Using Call Detail Records," 5 Sep. 2020. [Online]. Available: [URL if available].

3. P. Jain, "Automatic Traffic Signal Controller for Roads by Exploiting Fuzzy Logic," in *Proc. Int. Conf. Advances Commun. Network Comput.*, Berlin, Germany, 2011.

4. T. V. Janahiraman and M. S. M. Subuhan, "Traffic Light Detection Using Tensorflow Object Detection Framework," in *Proc. IEEE Int. Conf. System Eng. Technol.*, 2019.

5. A. Charef, Z. Jarir, and M. Quafafou, "Smart System for Emergency Traffic Recommendations: Urban Ambulance Mobility," *Int. J. Adv. Computer. Sci. Appl.*, vol. 13, no. 10, pp. 1-8, 2022.

6. F. I. Sapundzhi and M. S. Pop Stoilov, "Optimization algorithms for finding the shortest paths," *Bulgarian Chem. Commun.*, vol. 50, Special Issue B, pp. 115-120, 2018.

7. D. Sarpal, Y. Asthana, and M. K. Hota, "Review on Smart Traffic Management System for Ambulance," *Int. J. Electr. Eng. Technol.*, vol. 11, no. 6, Article ID: IJEET_11_06_001, 2020.

8. P. Devi and S. Anila, "Intelligent Ambulance with Automatic Traffic Control," in *Proc. 2020 Int. Conf. Computing Inf. Technol. (ICCIT-1441)*, Tabuk, Saudi Arabia, 2020, pp. 1-4, doi: 10.1109/ICCIT-144147971.2020.9213796.

9. Y. Mahima, T. Ginige, and R. Abeygunawardana, "Dynamic Traffic Light Controlling System Using Google Maps and IoT," 2020.

10. A. Sharma and R. Singh, "Reinforcement Learning Traffic Signal Control Method Based on Traffic Intensity Analysis," *IEEE Access*, vol. 9, pp. 45-55, Mar. 2021.

11. S. Kim and B. Park, "Automation of Traffic Control Systems with Deep Q-Learning Network," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 3, pp. 1234-1242, Apr. 2021.

12. J. Zhou, W. Zhang, and Y. Liu, "Traffic Flow Monitoring System Based on Mobile Phone Signaling Data," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 6850-6859, Aug. 2020.

13. H. R. Patel, K. B. Roy, and P. Desai, "GPS Based Adaptive Traffic Light Timings and Lane Scheduling," *IEEE Trans. Veh. Technol.*, vol. 69, no. 5, pp. 5055-5062, May 2020.

14. L. G. Yang and F. C. Wang, "Study of Automatic Traffic Signal System Using Inductive Loop Detectors," *IEEE Access*, vol. 8, pp. 12456-12465, Jan. 2020.

15. A. Gupta and S. Kumar, "Analysis of Traffic Flow and Optimization Using Neural Networks," *IEEE Access*, vol. 8, pp. 10234-10245, Feb. 2020, doi: 10.1109/ACCESS.2020.2965000.

Appendix

a. List of Figures

b. List of tables

Research paper:

Smart Urban Traffic Optimization for Emergency Services & Prioritization System

Mrs.Pallavi Saindane[1$],Atharva Hande [1], Aryan Surve[1], Tarunkumar Gatla [1], Sanchet Khemani [1]
[1] Vivekanand Education Society's Institute of Technology, Chembur, Mumbai 400 074. India

Atharva Hande
Department of – Computer Engineering
Vivekanand Education Society's Institute of Technology
Mumbai, India
d2022.atharva.hande@ves.ac.in

Tarunkumar Gatla
Department of – Computer Engineering
Vivekanand Education Society's Institute of Technology
Mumbai, India
d2022.tarunkumar.gatla@ves.ac.in

Aryan Suvre
Department of – Computer Engineering
Vivekanand Education Society's Institute of Technology
Mumbai, India
d2022.aryan.surve@ves.ac .in

Sanchet Khemani
Department of – Computer Engineering
Vivekanand Education Society's Institute of Technology
Mumbai, India
2022.sanchet.khemani@ves.ac.in

Abstract : Traffic congestion in cities often leads to delays, especially for emergency vehicles like ambulances, which need a clear path to save lives. Our Smart Urban Traffic Optimization for Emergency Services & Prioritization System aims to tackle this issue by using video processing and smart traffic signals.

This paper presents Smart Urban Traffic Optimization for Emergency Services & Prioritization System, an intelligent traffic management solution that leverages real-time video processing and adaptive signal control to enhance traffic flow efficiency. The system employs computer vision techniques to analyze vehicle density in each lane, dynamically adjusting traffic signals to prioritize highly congested routes. More critically, emergency vehicles are detected through advanced image processing, and immediate right-of-way is granted by overriding standard signal sequences, ensuring rapid transit.

The system analyzes vehicle density in each lane and ensures that the most crowded lane gets a green signal first. More importantly, if an emergency vehicle is detected, it is given the highest priority, allowing it to pass without delay.

Regular vehicles like cars and buses have a lower priority, ensuring a fair and efficient flow of traffic. The hardware component further enhances traffic management by automating signal control using smart algorithms. By combining real-time video analysis, intelligent decision-making, and automated signals, this system ensures faster emergency response times, reduces congestion, and makes urban roads safer for everyone.

Keywords : Smart Urban Traffic Optimization for Emergency Services & Prioritization System, Smart Traffic Management, Emergency Vehicle Prioritization ,Intelligent , Traffic Control, Urban Traffic Optimization, Real-Time Video Processing, Computer Vision in Traffic, Adaptive Traffic Signals

1. Introduction

Traffic congestion has become an unavoidable reality in modern cities, causing widespread delays, unnecessary fuel consumption, and elevated pollution levels. Beyond the daily frustration for commuters, congestion poses a significant risk to emergency services, such as ambulances, fire trucks, and police vehicles, which rely on rapid response times to save lives and minimize damage. The inability of these vehicles to navigate through traffic efficiently can lead to life-threatening delays, endangering public safety and straining emergency response systems.

Current traffic management strategies primarily depend on fixed signal timings, which fail to adjust dynamically to real-time traffic patterns. This rigid approach often results in inefficient vehicle movement, contributing to prolonged gridlocks and increased travel times. Additionally, the absence of an intelligent mechanism to identify and prioritize emergency vehicles exacerbates the problem, making it increasingly difficult to provide them with the swift passage they require. Addressing these challenges demands a smarter, technology-driven approach that optimizes traffic flow while ensuring emergency services receive top priority.

This research introduces the Smart Urban Traffic Optimization for Emergency Services & Prioritization System, an innovative solution designed to transform urban traffic management by integrating real-time video analysis and adaptive traffic signals. Utilizing state-of-the-art computer vision and AI-driven algorithms, the system continuously monitors vehicle density in each lane, ensuring that the most congested lane is given precedence in signal changes. More importantly, the system is capable of detecting emergency vehicles and instantly granting them the highest priority, ensuring they move through intersections without unnecessary delays. By leveraging this automated approach, the system drastically reduces response times for critical services, ultimately enhancing public safety and efficiency.

By seamlessly combining real-time traffic analysis, intelligent decision-making, and automated signal adjustments, our proposed solution ensures a fair and streamlined traffic flow for all vehicles while giving utmost importance to emergency services. The system not only improves road safety and efficiency but also aligns with the vision of smarter, more sustainable urban infrastructure. This research paper delves into the development, implementation, and real-world impact of our system, demonstrating its potential to revolutionize urban mobility and redefine modern traffic control strategies.

2. Literature Survey

Computer Vision and YOLO for Traffic Management

O. Tabajou's research explores the integration of Convolutional Neural Networks (CNNs) and the YOLO (You Only Look Once) algorithm for real-time traffic management. CNNs are widely used in computer vision applications to process video feeds from traffic cameras, allowing them to accurately identify vehicles, pedestrians, and cyclists. YOLO, a highly efficient object detection algorithm, is utilized to detect and locate these objects in real-time, ensuring traffic signals can dynamically adjust based on the density of vehicles in each lane. Similarly, M. Humayun, M. F. Almufareh, and N. Z. Jhanjhi emphasize the significance of autonomous systems in managing emergency vehicles through advanced visual sensing techniques. By implementing these technologies, urban traffic management can be significantly improved, providing a more responsive and efficient system.

However, despite its advantages, computer vision-based traffic management systems face several challenges. Poor lighting conditions, adverse weather, and environmental factors can affect the accuracy of object detection, as noted by K. Nellore and G. P. Hancke. Furthermore, these systems often rely on high-resolution video feeds, which raise concerns regarding privacy and computational efficiency. To address these challenges, researchers like B. Sun et al. have proposed the use of lightweight neural networks such as MobileNet and federated learning techniques, which enhance system scalability and responsiveness while reducing the computational burden and privacy risks associated with high-resolution video processing.

Comparative Analysis of RL-Based and Computer Vision Systems

Reinforcement Learning (RL)-based systems offer an alternative approach to traffic signal optimization. A. Sharma and R. Singh have proposed the use of Q-learning algorithms that continuously adapt traffic signal timings based on real-time traffic data. These systems learn from historical patterns and optimize signal changes to minimize delays. However, as highlighted by S. Kim and B. Park, RL-based approaches require extensive training periods and significant computational resources, making them less efficient for immediate real-world implementation.

Their study on deep Q-learning networks demonstrates how RL systems can reduce congestion, but they may struggle with sudden traffic fluctuations.

Conversely, computer vision-based solutions, such as those proposed by O. Tabajou, offer immediate feedback by processing live video feeds. While these systems are effective in dynamically adjusting signals, they are also susceptible to environmental disturbances, as discussed by K. Nellore and G. P. Hancke. S. K. R. Sanjana et al. further reinforce the advantages of visual-based approaches but acknowledge their limitations in adverse conditions. This comparative analysis underscores the need for a hybrid system that combines the adaptability of RL with the real-time efficiency of computer vision-based solutions.

IoT and Emergency Vehicle Management

IoT (Internet of Things) technology has gained traction as an effective solution for modern traffic management. Several researchers, including M. Humayun et al. and S. K. R. Sanjana et al., propose hybrid models that integrate computer vision with IoT-based sensors to prioritize emergency vehicles. These systems dynamically adjust traffic lights, ensuring emergency vehicles can pass through intersections without unnecessary delays. A. Gupta and S. Kumar further extend this idea by incorporating neural networks with IoT-based traffic monitoring, optimizing signal plans in real time by analyzing traffic patterns.

Other studies focus on GPS-based lane scheduling and signal timing management. H. R. Patel, K. B. Roy, and P. Desai describe an adaptive system that utilizes GPS tracking to manage traffic flow, prioritizing emergency vehicles without disrupting overall traffic movement. Additionally, Y. Mahima et al. leverage data from Google Maps and IoT sensors to dynamically control traffic signals, enhancing both emergency response times and urban mobility. These IoT-based approaches highlight the transformative potential of integrating smart technology with traditional traffic systems to create a more adaptive and efficient urban infrastructure.

Advancements in Computer Vision for Traffic Systems

A recent study titled *An Improved Smart Traffic Signal using Computer Vision* explores how CNNs and YOLO can be leveraged to improve traffic management. By processing video feeds, CNNs detect various traffic elements with high accuracy, while YOLO ensures fast object detection within each frame. This enables real-time signal adjustments based on actual traffic density.

One of the key benefits of computer vision-based traffic systems is their ability to adapt to real-time conditions without relying on predefined traffic rules. This flexibility is essential in modern urban settings, where traffic patterns can shift rapidly due to road work, accidents, or peak-hour congestion. For instance, when a road experiences heavy traffic, the system can extend the green light duration to ease congestion. Conversely, if pedestrian movement is high, the system can prioritize pedestrian safety by adjusting signal timings accordingly.

Despite these advantages, challenges remain. Computer vision-based traffic systems depend on high-quality video feeds, which can be affected by poor lighting, fog, or extreme weather conditions. This can impact the accuracy of detection models, leading to suboptimal traffic management. Additionally, processing high-resolution video feeds from multiple cameras in real time demands significant computational resources, increasing both deployment costs and complexity. Privacy concerns also arise with widespread video surveillance.

To overcome these limitations, researchers suggest integrating lightweight neural networks like MobileNet and employing federated learning techniques. These advancements can help reduce the computational burden while maintaining the efficiency of real-time traffic monitoring. By addressing these challenges, future traffic systems can achieve a balance between responsiveness, scalability, and privacy protection, making them a viable solution for smart cities.

This literature survey highlights the strengths and limitations of various approaches, emphasizing the need for hybrid models that incorporate the best aspects of RL, computer vision, and IoT for a holistic traffic management solution.

3. Proposed System Architecture

YOLOv8 detects emergency vehicles in real-time video streams by following a structured workflow that ensures high accuracy and efficiency. (1) Video Input & Frame Extraction: The system captures video input from sources like traffic surveillance cameras, dashcams, drones, or live webcams. The video is read frame by frame using OpenCV, converting it into a sequence of images for processing. The frame rate (e.g., 30 FPS) is maintained to ensure smooth detection. Each extracted frame is resized to a standard resolution (typically 640×640 pixels) to match YOLOv8's input format.

This resizing ensures consistency and speeds up model inference. (2) Preprocessing & Feature Extraction: The extracted frames undergo preprocessing, including contrast enhancement, normalization, and noise reduction, to improve detection under varying lighting conditions. CSPDarkNet, the backbone of YOLOv8, extracts key visual features like shape, texture, and edges of vehicles. The Feature Pyramid Network (BiFPN) further enhances detection by analyzing objects at different scales, enabling the model to detect both near and far emergency vehicles in a scene. (3) Object Detection & Classification: YOLOv8 employs an anchor-free detection mechanism, which dynamically predicts bounding boxes instead of using predefined anchor boxes. This improves localization accuracy, particularly for vehicles of different sizes. The model assigns a confidence score (ranging from 0 to 1) to each detected object, determining whether it belongs to an emergency vehicle category (e.g., ambulance, fire truck, police car).

A classification layer then identifies the specific type of vehicle. (4) Post-Processing & Visualization: To avoid redundant detections, non-maximum suppression (NMS) is applied, removing overlapping bounding boxes and retaining only the most relevant ones. The detected emergency vehicles are then highlighted with rectangular bounding boxes, and class labels such as "Ambulance: 0.92" are displayed on the video frame. (5) Real-Time Decision Making: The detection output can be utilized for traffic management systems to prioritize emergency vehicles by adjusting traffic signals dynamically. In autonomous vehicles, the system can enable automatic braking or lane switching to yield to emergency vehicles. Law enforcement agencies can integrate this technology into smart surveillance systems for real-time monitoring and response. (6) Advantages: YOLOv8 provides real-time inference, ensuring quick and accurate emergency vehicle detection. Its deep learning-based architecture ensures high accuracy even in low-light, foggy, or crowded conditions. The model is scalable and can be deployed on edge devices, traffic cameras, or autonomous vehicle systems. (7) By leveraging advanced feature extraction and real-time object detection, YOLOv8 efficiently detects emergency vehicles, playing a crucial role in enhancing traffic management, autonomous driving safety, and public emergency response efficiency.

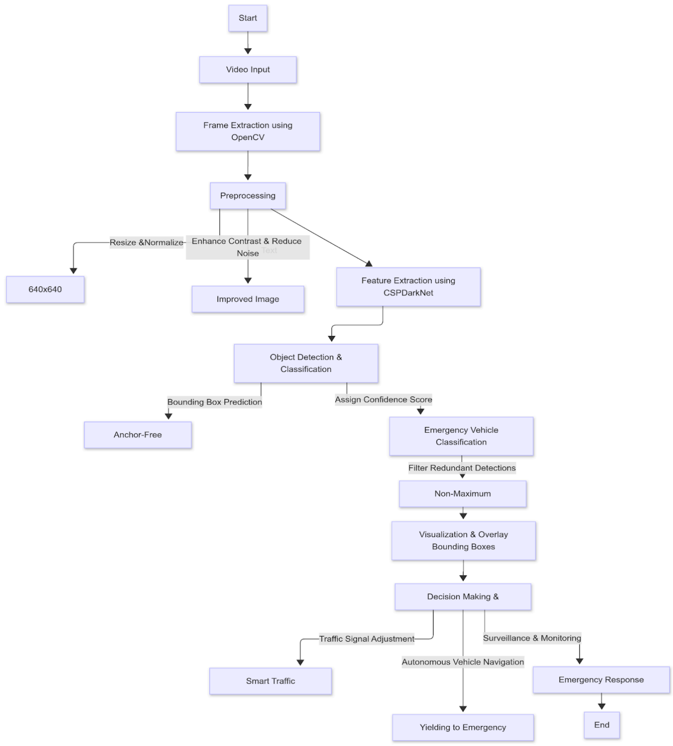


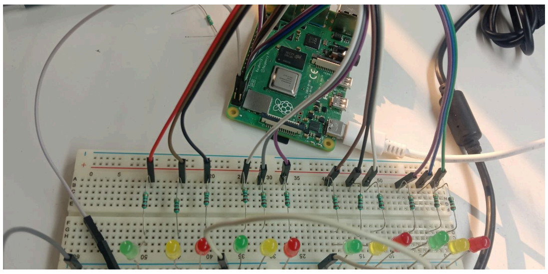
Fig .2. YOLOv8-based emergency vehicle detection system

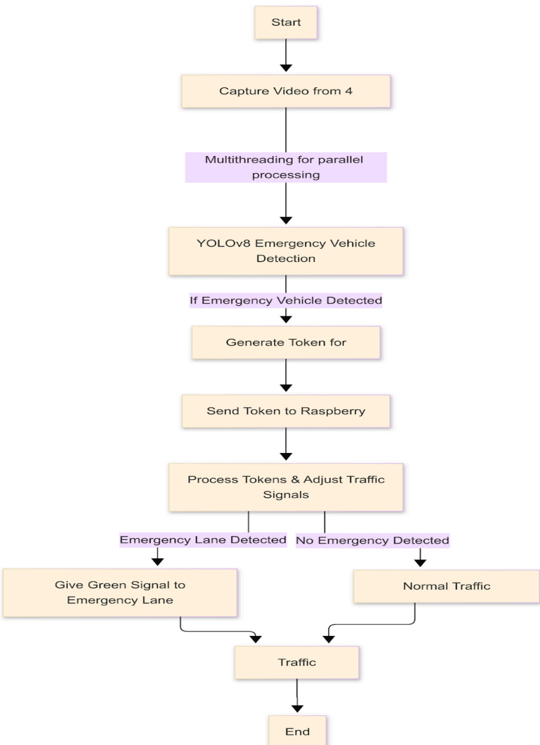
4. Experimental Setup and Implementation details

The hardware system seamlessly integrates with the YOLOv8-based software to enable dynamic traffic signal control using Raspberry Pi and a token-based priority mechanism. (1) Software Detection & Token Generation: The YOLOv8 model processes real-time video feeds from four traffic lanes, detecting emergency vehicles such as ambulances, fire trucks, and police cars. For every detected emergency vehicle, the system generates a token, associating it with the specific lane where the vehicle was found.

The token contains details like lane number, detection confidence, and timestamp to ensure precise decision-making. (2) Token Transmission to Raspberry Pi: The generated tokens are transmitted to the Raspberry Pi via a communication protocol such as MQTT, HTTP, or Serial UART. The Raspberry Pi continuously monitors incoming tokens and prioritizes lanes dynamically, ensuring that emergency vehicles get a higher priority over regular traffic.

Once the Raspberry Pi receives the tokens, it executes signal processing and traffic control operations. (3) Decision-Making & Traffic Light Control: Based on the lane with the highest priority (i.e., the lane where an emergency vehicle is detected), the Raspberry Pi sends GPIO signals to control relay modules or LED-based traffic lights, switching the corresponding lane's signal to green while keeping other lanes on hold. (4) Dynamic Traffic Adjustment: As soon as the emergency vehicle clears the intersection, the Raspberry Pi removes the token from memory and restores normal signal sequencing. If multiple emergency vehicles are detected in different lanes, the Raspberry Pi queues tokens in order of priority and manages traffic without manual intervention. (5) Key Advantages: This system enables real-time traffic optimization, faster emergency response, and adaptive lane management, reducing congestion and ensuring smooth movement of emergency vehicles.





References

1. A. N. A. Yusuf, A. S. Arifin, and F. Y. Zulkifli, "Recent Development of Smart Traffic Lights," *Int. J. Artif. Intell.*, vol. 10, no. 1, pp. 224-233, Mar. 2021.

2. O. Tabajou, "Intelligent Traffic Light System by Using Video Processing," *2nd Int. Conf. Eng. Nat. Soc. Sci.*, Apr. 2023.

3. M. Humayun, M. F. Almufareh, and N. Z. Jhanjhi, "Autonomous Traffic System for Emergency Vehicles," *Electronics*, vol. 11, no. 4, Feb. 2022.

4. P. Devi and S. Anila, "Intelligent Ambulance with Automatic Traffic Control," in *Proc. 2020 Int. Conf. Computing Inf. Technol. (ICCIT-1441)*, Tabuk, Saudi Arabia, 2020, pp. 1-4, doi: od10.1109/ICCIT-144147971.2020.9213796.

5. J. Zhou, W. Zhang, and Y. Liu, "Traffic Flow Monitoring System Based on Mobile Phone Signaling Data," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 6850-6859, Aug. 2020.

6. P. Patel and R. Verma, "Traffic Management Using Machine Learning Techniques," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 4, pp. 3456-3467, Apr. 2021, doi: 10.1109/TITS.2020.3045001.

7. M. A. Hossain, A. Rahman, and M. M. Alam, "Dynamic Traffic Light Controlling System Using Google Maps and IoT," *Int. J. Sci. Res. Sci. Technol.*, vol. 8, no. 5, pp. 250-257, Dec. 2020.

8. D. Sarpal, Y. Asthana, and M. K. Hota, "Review on Smart Traffic Management System for Ambulance," *Int. J. Electr. Eng. Technol.*, vol. 11, no. 6, Article ID: IJEET_11_06_001, 2020.

9. K. Nellore and G. Hancke, "Traffic Management for Emergency Vehicles Using IoT and Visual Sensing," *IEEE Sensors*, vol. 16, pp. 1892, 2016.